

HOANG XUAN PHI LONG

SOFTWARE ENGINEER/ AI INTERN

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SUMMARY

Second-year Software Engineering student at HUIT with a strong dual focus on systems programming and AI/ML research. Awarded 2nd Place at the Faculty-level and an Honorable Mention at the University-level Scientific Research Competition for building a fully local Vietnamese conversational AI system (STT · LLM · RAG · TTS). Actively seeking a part-time Software Engineering or AI/ML internship to apply research and development skills in a professional setting.

EDUCATION

Ho Chi Minh City University of Industry and Trade (HUIT)

Aug 2024 - Present

B.Eng. in Information Technology - Software Engineering

GPA: 3.76 / 4.0 (8.93/10) · Merit Scholarship (Semester I & II, 2024–2025) · Excellence Scholarship (Semester I, 2025–2026) · Distinguished Student Award 2024–2025

AWARDS & COMPETITIONS

Research Awards

- 2nd Place — Faculty-level Scientific Research Competition, "Local Vietnamese Conversational AI System" (2026)
- Honorable Mention — University-level Scientific Research Competition, "Local Vietnamese Conversational AI System" (2026)
- Honorable Mention — Faculty Digital Transformation Challenge, "Diabetes Diagnosis via Machine Learning" (2026)

Competitive Programming

- ICPC National 2025
- Vietnam Informatics Olympic (OLP.TIN), 34th edition
- Provincial Youth Informatics Competition, Quang Tri (2022)
- Consolation Prize & Top Preliminary Scorer | "Algorithm Arena" Competition – Faculty of IT, HUIT (2026)

PROJECTS

LOCAL VIETNAMESE CONVERSATIONAL AI SYSTEM

Jan 2026 – May 2026

2nd Place – Faculty Research Competition | Honorable Mention – University Research Competition

Tech: Python · Whisper · Wav2Vec 2.0 · Sherpa-ONNX · ONNX Runtime · PyTorch · Qdrant · RAG

Role: Speech-to-Text (STT) Module Lead | Team Research Project

- Surveyed and benchmarked 3 **STT architectures** for Vietnamese: End-to-End Transformer (Whisper), CTC-based (Wav2Vec 2.0), and Zipformer (Sherpa-ONNX), evaluated on WER, CER, and Real-Time Factor (RTF) across Google FLEURS, Mozilla Common Voice 24.0, and VLSP 2020 - Wav2Vec 2.0 achieved **WER 22.7%** on FLEURS with **RTF = 0.007**
- Optimized CPU-only inference by converting models to ONNX format and applying INT8 Post-Training Quantization, achieving RTF < 1.0 and reducing latency by ~50% vs. FP32
- Designed the STT module's integration interface within the full Speech-to-Speech pipeline (STT → LLM+RAG → TTS) using a modular architecture enabling independent component replacement

SKIN CANCER DIAGNOSIS - HYBRID CNN-TRANSFORMER

Aug 2025 – Mar 2026

Tech: Python · PyTorch · ResNet-50 · Swin Transformer · EfficientNet · ViT · ISIC 2019

Role: Research Member (Model Training & Evaluation) | Team Research Project

- Built and trained 4 **Hybrid CNN-Transformer architectures** using Feature Fusion (concatenating CNN local features + Transformer global representations) on the ISIC 2019 multi-class skin lesion dataset
- Best model - **ResNet-50 + Swin Transformer** achieved **Accuracy 91.06%**, Precision 0.9108, Recall 0.9106, F1 0.9107 after 30-epoch fine-tuning from ImageNet pre-trained weights (Adam · ReduceLROnPlateau · Binary Cross-Entropy)
- Analyzed Confusion Matrices for all 4 models; ResNet+Swin recorded the **lowest False Negative rate** - the clinically critical metric for cancer screening applications

PRIVATE CLINIC MANAGEMENT SYSTEM

Sep 2025 – May 2026

Tech: C# · WPF · MVVM · Entity Framework · SQL Server · LINQ · LiveCharts.Wpf · EPPlus

Role: Team Lead & Core Developer | .NET Technology Coursework

- Designed a **12-table SQL Server schema** and implemented **Role-Based Access Control (RBAC)** dynamically rendering navigation menus for 5 roles (Admin, Doctor, Receptionist, Pharmacist, Cashier) via WPF BooleanToVisibilityConverter
- Architected the full project using **MVVM pattern** with INotifyPropertyChanged and RelayCommand, completely decoupling UI from business logic
- Built the end-to-end clinical workflow: patient intake → diagnosis & prescription → **automatic inventory deduction (EF Transaction)** → invoice generation
- Developed Admin statistics dashboard using **LINQ aggregation** (.Sum(), .Count(), GroupBy) with monthly revenue bar charts (LiveCharts.Wpf) and Excel export (EPPlus, no Office required)

Github: github.com/YuluWusu/DoAn_QuanLyBenhVien

ONLINE MOVIE TICKET BOOKING WEBSITE

Sep 2025 – May 2026

Tech: ASP.NET MVC · Entity Framework 6 · SQL Server · jQuery · AJAX · Bootstrap 5

Role: Team Lead & Full-stack Developer | Web Programming Coursework

- Designed an **11-table relational schema** and built the full customer portal: registration/login (duplicate phone number/ Email/Personal ID validation), movie listing with status filters, quick search, and showtime display
- Built Admin portal: movie CRUD with poster upload, cinema/room management with **auto-generated seat maps**, and showtime scheduling with **conflict detection** to prevent overlapping bookings
- Applied **AJAX** for asynchronous seat-map and showtime loading; real-time price calculation by seat type (VIP/ Standard) without page reload

Github: github.com/YuluWusu/DoAn_QuanLyVeXemPhimTrucTiep

SKILLS

- **Languages:** Python, C#, C++ (OOP), SQL
- **AI / ML / Data:** PyTorch, Whisper, Wav2Vec 2.0, Sherpa-ONNX, ONNX Runtime, Swin Transformer, ResNet-50, ViT, EfficientNet, RAG, Qdrant, NLP / Speech-to-Text, CNN, Deep Learning, Data Preprocessing
- **Software & Web:** ASP.NET MVC, Entity Framework 6, WPF / XAML / MVVM, SQL Server, jQuery, AJAX, Bootstrap 5, HTML/CSS, Database Design
- **Languages & Certifications:** TOEIC 850 · AI Agent Programming Certificate (Data Processing & Code Generation)
- **Soft Skills:** Research writing & presentation, Teamwork, Problem-solving